

Lijun Zhang

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CURRENT POSITION

Zhejiang University of Technology, Hangzhou, Zhejiang, China
Assistant Professor in Computer Science

Aug.2026 – Present

EDUCATION

University of Massachusetts, Amherst, Amherst MA, USA
Ph.D. in the College of Computer Science

Sept.2019 – May. 2025
Overall GPA: 4.0/4.0

Tongji University, Shanghai, China
M.Sc. in Software Engineering

Sept.2016 - Mar.2019
Overall GPA: 4.63/5.0

Tongji University, Shanghai, China
B.Eng. in Software Engineering

Sept.2012 - Jun.2016
Overall GPA: 4.78/5.0

WORK EXPERIENCES

Amazon Robotics, Seattle, WA, USA
Postdoctoral Scientist

June.2025 – June.2026

Summary: Conduct independent research bridging cutting-edge machine learning techniques with practical system-level applications.

Dolby Laboratories, Sunnyvale, CA, USA
PhD Research Intern

May.2024-August.2024

Summary: Create a universal solution to image restoration tasks with pre-trained diffusion models in a zero-shot manner.

- Designed an unsupervised learning-based image restoration framework to recover a given distorted image, e.g., blurry images or low-light images, using pre-trained image generation diffusion models.
- Restored details in distorted images even with complex distortion types, such as low-light blurry images, with real time performance.

Amazon Robotics, Seattle, WA, USA
Applied Scientist II Summer Intern

May.2023-August.2023

Summary: Design and provide multi-task solution to the multiple vision tasks involved in the Amazon Robotics Stow system.

- Designed a multi-task framework that can solve semantic segmentation, instance segmentation, depth estimation, and surface normal prediction simultaneously.
- Overcame the challenge from disjoint training datasets for different tasks and successfully constructed a multi-task model with high accuracy as single-task models and much lower latency and computations.

RESEARCH INTERESTS

My research interests lie in *Automated Machine Learning Engineering and Computer Vision*. My research includes exploring the generative capability of diffusion models beyond image generation, and developing multi-task models that achieve high task accuracy, small memory footprint, and low computation cost simultaneously for vision tasks. My previous works have spanned low-light image enhancement, image resolution, and image quality assessment.

PUBLICATIONS

- Xiao Liu, **Lijun Zhang**, Deepak Ganesan, Hui Guan, [Communication-Efficient Multi-Device Inference Acceleration for Transformer Models](#), in International Conference on Machine Learning (ICML), 2026.
- **Lijun Zhang**, Xiao Liu, Kaleel Mahmood, Caiwen Ding, Hui Guan, Attacking all tasks at once using adversarial examples in multi-task learning, in Neurocomputing, 2025
- **Lijun Zhang**, Xiao Liu, Hui Guan, [Reimagining Parameter Space Exploration with Diffusion Models](#), in EXAIT workshop at International Conference on Machine Learning (ICML), 2025.
- **Lijun Zhang**, Xiao Liu, Antoni Viros Martin, Cindy Xiong Bearfield, Yuriy Brun, Hui Guan, [Attack-Resilient Image Watermarking Using Stable Diffusion](#), in Neural Information Processing Systems (NeurIPS), 2024.
- Kunjal Panchal, Nisarg Parikh, Sunav Choudhary, **Lijun Zhang**, Yuriy Brun, Hui Guan, [Thinking Forward: Memory-Efficient Federated Finetuning of Language Models](#), in Neural Information Processing Systems (NeurIPS), 2024.
- Xiao Liu, **Lijun Zhang**, Hui Guan, [Information-Enhanced Graph Neural Network for Transcending Homophily Barriers](#), IEEE Access 2024.
- Qizheng Yang, Tianyi Yang, Mingcan Xiang, **Lijun Zhang**, Haoliang Wang, Marco Serafini, Hui Guan, [GMorph: Accelerating Multi-DNN Inference via Model Fusion](#), European Conference on Computer Systems (EuroSys), 2024.
- Kunjal Panchal, Sunav Choudhary, Nisarg Parikh, **Lijun Zhang**, Hui Guan, [Flow: Per-instance Personalized Federated Learning](#), in Conference on Neural Information Processing Systems (NeurIPS), 2023.
- **Lijun Zhang**, Xiao Liu, Hui Guan, [A Tree-Structured Multitask Model Architectures Recommendation System](#), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2023
- **Lijun Zhang**, Qizheng Yang, Xiao Liu, Hui Guan, [An Alternative Hard-Parameter Sharing Paradigm for Multi-Domain Learning](#), IEEE Access, 2023.
- **Lijun Zhang**, Xiao Liu, Hui Guan, [AutoMTL: A Programming Framework for Automating Efficient Multi-Task Learning](#), in Conference on Neural Information Processing Systems (NeurIPS), 2022.
- **Lijun Zhang**, Xiao Liu, Hui Guan, [A Tree-Structured Multi-Task Model Recommender](#), in International Conference on Automated Machine Learning (AutoML), 2022.
- **Lijun Zhang**, Qizheng Yang, Xiao Liu, Hui Guan, [Rethinking Hard-Parameter Sharing in Multi-Domain Learning](#), in IEEE International Conference on Multimedia and Expo (ICME), 2022.
- **Lijun Zhang**, Hui Guan, Yufei Ding, Xipeng Shen, Hamid Krim, [Reuse-centric K-means Configuration](#), in Information Systems, 2021.
- Hui Guan, Umang Chaudhary, Yuanchao Xu, Lin Ning, **Lijun Zhang**, and Xipeng Shen, [Recurrent Neural Networks Meet Context-Free Grammar: Two Birds with One Stone](#), in IEEE International Conference on Data Mining (ICDM), 2021.
- Lin Zhang, **Lijun Zhang**, Xiao Liu, Ying Shen, Shengjie Zhao, Shaoming Zhang, [Zero-Shot](#)

[Restoration of Back-lit Images Using Deep Internal Learning](#), in ACM International Conference on Multimedia (ACM MM), 2019.

- **Lijun Zhang**, Lin Zhang, Xiao Liu, Ying Shen, and Dongqing Wang, [Image exposure assessment: A benchmark and a deep convolutional neural networks based model](#), in IEEE International Conference on Multimedia and Expo (ICME), 2018.
- **Lijun Zhang**, Lin Zhang, and Lida Li, [Illumination quality assessment for face images: a benchmark and a convolutional neural networks based model](#), in International Conference on Neural Information Processing (ICONIP), 2017.

PREPRINTS

- **Lijun Zhang**, Ruinian Xu, Mudit Agrawal, [Simple Supervision Is Hard to Beat: A Bitter Lesson from Sparse Target Labels in Domain-Adaptive Object Detection](#), in ArXiv, 2026.
- **Lijun Zhang**, Nikhil Chacko, etc., [Visual Foresight for Robotic Stow: A Diffusion-Based World Model from Sparse Snapshots](#), in ArXiv, 2026.
- **Lijun Zhang**, Xiao Liu, Erik Learned-Miller, Hui Guan, [SID-NISM: A Self-supervised Low-light Image Enhancement Framework](#), in ArXiv, 2020.

RESEARCH EXPERIENCES

Exploring Stable Diffusion Models

Sept.2023-June.2026

Research Assistant, Advised by Prof. Hui Guan, MLSys Lab, UMass Amherst

Summary: Rapid evolution in deep generative models has led to methods capable of synthesizing high-quality, realistic images. As a representative, stable diffusion models become popular in both traditional and emerging computer vision tasks. We would like to explore whether powerful diffusion models could serve as an off-the-shelf tool for downstream tasks, and whether the philosophy of multi-task learning and the power of stable diffusion models could facilitate each other.

- Adapted well-trained stable diffusion models to image watermarking task without additional training process; our research demonstrates that stable diffusion is a promising approach to robust watermarking, able to withstand even stable-diffusion-based attacks. (A paper accepted to NeurIPS'24)
- Exploring possibility of efficiently generating parameters for multi-task models from single-task models, utilizing the generation power of stable diffusion models. (A paper accepted to EXAIT@ICML'25)
- Understanding the implicit modeling of the world achieved by the diffusion model during learning generation task, which helps to understand the physical laws of the world and build a world model with the ability to predict future states as human can (postdoc focus).

Programming Systems for Efficient Multi-Task Learning

Oct.2020-June.2025

Research Assistant, Advised by Prof. Hui Guan, MLSys Lab, UMass Amherst

Summary: A fundamental challenge in multi-task learning is to determine the set of parameters to share across tasks to achieve the best performance for tasks. Existing manually designed network architectures and learning-based methods have limitations on either task performance or application generality. There is a strong need to develop programming systems for efficient multi-task learning that overcome current issues.

- Conducted an empirical study on how to share model parameters in multi-domain learning and concluded insights that challenge the common practice in hard parameter sharing and promote an alternative parameter sharing strategy as a stronger baseline. (A paper published on ICME'22, IEEE Access'23)

- Designed a tree-structured multi-task models recommender that explore the architecture design space completely and automatically in a white-box manner via building recursive space enumerator and reliable task accuracy estimator; our recommended multi-task architectures are competitive with state-of-the-art under specified computation budgets. (A paper published on AutoML'22, TNNLS'23)
- Proposed a programming framework that largely automates architecture search of multi-task models given an arbitrary backbone model and a set of tasks via compiler support and policy-architecture co-training; the framework could identify compact multi-task models that outperform state-of-the-art approaches in task performance. (A paper accepted by NeurIPS'22)
- Analyzed the robustness of multi-task systems with existing single-task adversarial attacks and identified their limitations in ruining all the tasks at once; developed dynamic gradient balanced attack to efficiently attack all the tasks simultaneously; demonstrated the relationship between parameter sharing and adversarial vulnerability of multi-task models. (A paper accepted by Neurocomputing'25)

Zero-shot Low-light Image Enhancement

Jan.2018-June.2020

Research Assistant, Advised by Prof. Erik G. Learned-Miller, Vision Lab, UMass Amherst

Prof. Lin Zhang, Computer Vision Lab, Tongji University

Summary: To get rid of the restriction of using training data when conducting image restoration via neural networks, it is necessary to design self-supervised image enhancement approaches to restore the quality of any single back-lit image only relying on the visual information of the image itself.

- Proposed a self-supervised image decomposition network based on Retinex Theory, an image decomposition theory, which takes a low-light image and its histogram equalization image as only inputs for image illumination extraction and noise removal. (A preprint posted on ArXiv'20)
- Modeled the S-curve adjustment procedure in the back-lit image restoration with Markov Random Field (MRF) and proposed the first self-supervised learning solution to estimate the S-curve parameters that best fit the back-lit image; as an image-specific framework with low computation cost, the proposed method could be applied to video stream directly. (A paper published on ACMMM'19)

No-reference Image Quality Assessment

May.2018-Sept.2018

Research Assistant, Advised by Prof. Lin Zhang, Computer Vision Lab, Tongji University

Summary: No-reference image quality assessment aims to design computerized algorithms that can evaluate the quality of a given image without reference in consistency with the perception of human observers, which could be objective metrics for many image processing problems.

- Established an image exposure database containing real-scene images captured in various scenarios under seven exposure levels as well as artificial images generated via a proposed image illumination adjustment tool, based on which an algorithm towards automatic image exposure level assessment is designed as a baseline. (A paper published on ICME'18)
- Established a database of face images with various illumination patterns and proposed a CNN-based method to evaluate face images illumination quality in high consistency (SROCC 0.95) with human perception. (A paper published on ICONIP'17)

SERVICE

- ICLR 2027 Area Chair, TMLR Yearly Reviewer
- NeurIPS 2022-2026, ICML 2023-2026, ICLR 2023-2026, AAAI 2027 Reviewer
- CVPR 2024-2025, ICCV 2025, ECCV 2026 Reviewer

RECOGNITION

UMass CICS PhD Dissertation Writing Fellowship	UMass Amherst CICS, MA, 2024
IBM PhD Fellowship	IBM, 2023-2024
Scholar Award & Top Reviewer	NeurIPS, 2022
Travel Grant Award	Conf-AutoML, 2022
The Lori a. Clarke Scholarship	UMass Amherst CICS, MA, 2020
Best Undergraduate & Graduate Thesis	Tongji University, Shanghai, 2016 & 2019
Outstanding Graduates in Shanghai	Education Committee, Shanghai, 2016 & 2019
National Scholarship for Graduate Students	Ministry of Education, China, 2018